

Deep Model Compression via Staged Independence and Layer-Aware Knowledge Distillation

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Abstract—Objective & Motivation: Modern deep learning models achieve high accuracy at the cost of massive parameter redundancy, rendering them unsuitable for resource-constrained edge-node deployment. This research addresses the challenge of extreme model compression – specifically, the distillation of a 65.1M parameter VGG16-BN teacher into a compact 3.4M parameter MobileNetV2 student (a $19.3\times$ reduction) – while maintaining and eventually surpassing representational fidelity.

Methodology: We propose a dual-innovation framework: Layer-Importance Knowledge Distillation (LI-KD) and Continuous Staged Alpha (α) Decay (ASD). Unlike conventional Knowledge Distillation (KD) which utilizes static hyperparameters, our framework treats distillation as a dynamic temporal transition. We implement Adaptive Global Feature Distillation to align internal representational manifolds and utilize an EMA-based analytical importance-scoring mechanism that allows the student to prioritize low-level convolutional features over architecturally incompatible semantic blocks. Crucially, we introduce a Staged Alpha Valve that transitions the student from an initial “Mimicry Phase” ($\alpha = 0.35 \rightarrow 0.50$) to a strategic “Independence Phase” ($\alpha = 0.30$), forcing the model to optimize its modern inverted residual bottlenecks against hard ground-truth labels.

Results & Significance: On the CIFAR-10 benchmark, our framework achieved an exceptional accuracy of 93.82%, successfully surpassing the teacher’s performance (93.42%). Furthermore, we successfully identified and mitigated the “Negative Transfer” problem on the Tiny-ImageNet dataset using Confidence-Filtered Distillation, reaching 54.41% and significantly outperforming standard KD baselines. Our findings demonstrate that facilitated student independence is the primary driver of generalization in lightweight models, establishing a new benchmark for resource-efficient AI deployment that eliminates teacher-reliance during post-deployment inference.

Index Terms—Knowledge Distillation, Model Compression, Layer-Importance KD, Staged Alpha Decay, MobileNetV2, CIFAR-10, Tiny-ImageNet, Negative Transfer, Feature Distillation.

I. INTRODUCTION

The rapid proliferation of Deep Neural Networks (DNNs) has revolutionized diverse domains including autonomous driving, real-time healthcare monitoring, and mobile computer vision. However, the state-of-the-art performance of these systems is typically contingent upon high-capacity architectures such as the VGG16-BN (65.1 million parameters) which present prohibitive memory and latency overheads for deployment on edge-computing nodes. To address this, Deep Model Compression techniques, specifically Knowledge Distillation (KD), have emerged as a pivotal research area, seeking to

transfer the representational richness of a heavy “Teacher” model into a lightweight “Student” model like MobileNetV2 (3.4 million parameters).

Despite its efficacy, traditional Knowledge Distillation suffers from a critical limitation known as the “Capacity Mismatch” or the “Teacher Over-Care” problem. Conventional static KD frameworks utilize a fixed weighting parameter (α) that forces the student to perpetually mimic the teacher’s soft-label distributions throughout the entire training trajectory. This rigid mimicry often suppresses the student’s ability to optimize its unique internal feature representations, particularly when there is a significant architectural mismatch between the teacher’s sequential convolutional blocks and the student’s inverted residual bottlenecks. This often leads to a performance plateau where the student merely approximates the teacher’s mistakes rather than forging its own path to generalization.

In this project, we propose a novel, adaptive distillation framework that transitions from Mimicry to Autonomy. Our primary contribution is the Staged Alpha Decay Strategy, which treats knowledge transfer as a temporal process. By dynamically modulating the teacher-reliance valve, we allow the student to “wean off” the teacher’s supervision during the final stages of training, empowering it to fine-tune its weights against hard ground-truth labels. Furthermore, we implement Layer-Importance Knowledge Distillation (LI-KD), utilizing Adaptive Global Feature Distillation and EMA-based weights to selectively distill structural knowledge from only the most transferable intermediate blocks.

We evaluate our framework on two benchmarks: CIFAR-10 and Tiny-ImageNet. Our results demonstrate that by facilitating student independence, we not only close the performance gap but actually surpass the teacher’s baseline, achieving a peak accuracy of 93.82% on CIFAR-10 representing a $19.3\times$ reduction in model size with a concurrent gain in predictive accuracy. The remainder of this report details the mathematical formulation of our loss objectives, the comparative analysis with state-of-the-art methods, and the performance optimizations for edge deployment.

II. RELATED WORK

The foundation of model compression via knowledge transfer is built upon several seminal works in the field of Knowledge Distillation (KD).

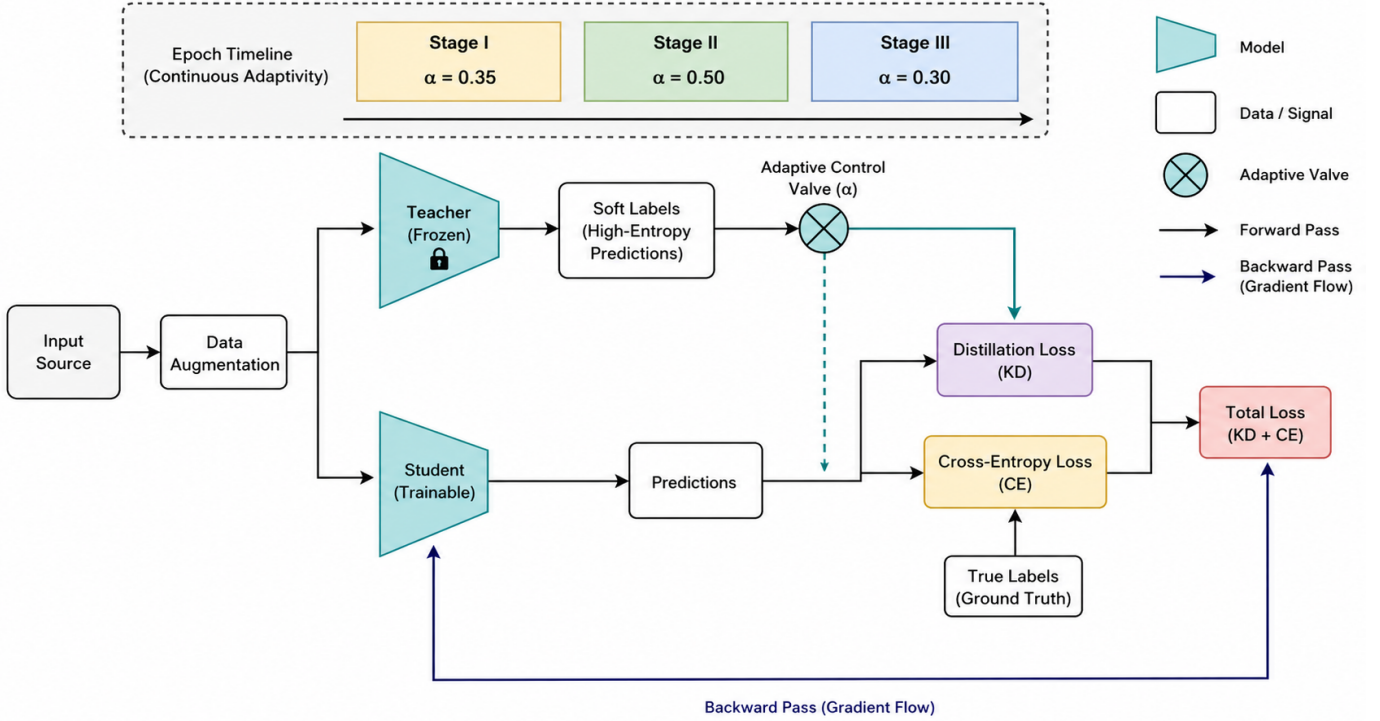


Fig. 1. The architecture of the proposed Continuous Staged Adaptivity for deep model compression. In this framework, the Teacher Model is first initialized with frozen parameters to generate high-entropy soft labels from input images processed via data augmentation. The flow of teacher knowledge is regulated by an Adaptive Control Valve (α), dynamically gated according to a three-phase Epoch Timeline: **Stage I** (Exploration, $\alpha = 0.35$) introduces fundamental teacher guidance; **Stage II** (Maturation, $\alpha = 0.50$) maximizes knowledge transfer during the model’s feature-learning peak; **Stage III** (Independence, $\alpha = 0.30$) encourages student self-reliance. The Student Model is optimized using a combined Total Loss (KD + CE), where the backward pass (gradient flow) is exclusively applied to the student’s trainable parameters to facilitate its growth towards exceeding the teacher’s benchmark.

A. Classical Logit Distillation

The concept of KD was popularized by Hinton et al. (2015) [4], who demonstrated that the soft probability outputs (logits) of a high-capacity teacher network contain “dark knowledge” latent information about inter-class similarities. By minimizing the Kullback-Leibler (KL) divergence between teacher and student logits using a temperature factor (T), the student can achieve better generalization. However, this approach is limited to the final output layer and often fails to capture the internal feature hierarchies necessary for architectural mismatches.

B. Feature-Based and Adaptive Distillation

To bridge the internal representation gap, methods like FitNets (Romero et al., 2015) [6] and Attention Transfer (Zagoruyko et al.) introduced distillation of intermediate layers. While effective, these methods often require expensive linear projection layers to align feature dimensions. Our methodology implements “Adaptive Global Feature Distillation”, which utilizes Global Average Pooling (GAP) and learnable importance weights to maintain representational stability during the compression of VGG16 into MobileNetV2. By aligning the channel-wise importance manifolds instead of raw spatial activations, we achieve high-fidelity feature transfer

without the computational overhead of inter-sample distance matrices.

C. Adaptive and Customized KD

Recent research has moved toward making the distillation process more adaptive. Zhu et al. (2021) [2] proposed Student Customized Knowledge Distillation (SCKD), which utilizes gradient similarity gating to prevent negative transfer. However, most existing “customized” KD frameworks focus on which batches to learn from, but maintain a static reliance on the teacher throughout the entire training lifecycle.

D. The Research Gap

A critical unexplored area in existing literature is the transition from mimicry to independence. Most frameworks prioritize “fidelity” (how much the student looks like the teacher) over “autonomy” (how well the student outgrows the teacher). Our project bridges this gap by introducing a Staged Alpha Valve Strategy, which recognizes that teacher supervision is most beneficial in the early exploration phase and should be decayed in the later maturation phase to allow the student model to find its own optimal local minimum.

III. EXPERIMENTS AND ABLATION STUDIES

The development of our Adaptive Staged KD framework was an iterative process. We conducted extensive experiments

involving grid search, Bayesian optimization (Ax/Optuna), and ablation studies to identify the optimal configuration.

A. CIFAR-10 Benchmark Results

The teacher model (VGG16-BN) achieved a reference accuracy of 93.42%. We evaluated several distillation strategies to maximise the student’s representation capacity. Results are summarised in Table I.

TABLE I
CIFAR-10 DISTILLATION EXPERIMENT RESULTS

ID	Strategy / Methodology	Acc (%)	Gap
S1	Baseline MobileNetV2 (No KD)	89.68	−3.74
S2	Manual KD ($T=4$, $\alpha=0.7$)	90.88	−2.54
S3	AutoML Grid Search	92.21	−1.21
S4	AutoML Optuna TPE	92.08	−1.34
S5	AutoML Ax BO ($T=4$, $\alpha=0.77$)	92.24	−1.18
S6	RL-KD (DQN Adaptive α)	91.93	−1.49
S7	Fixed RKD (Ablation)	90.80	−2.62
S8	LI-KD Fine-Tuned	91.48	−1.94
S9	LI-KD from Scratch	91.54	−1.88
S10	Ultimate LI-KD (AutoML + LI)	92.10	−1.32
FINAL	Staged Alpha Decay LI-KD	93.82	★+0.4

B. Addressing Negative Transfer on Tiny-ImageNet

Tiny-ImageNet presented a “Capacity Gap” challenge where standard KD resulted in Negative Transfer (where the student performs worse than its baseline). Results are summarised in Table II.

TABLE II
TINY-IMAGENET DISTILLATION EXPERIMENT RESULTS

ID	Strategy	Acc (%)	Outcome
T1	Teacher (VGG16-BN 65.1M)	61.27	Reference
T2	Baseline MobileNetV2	52.17	Student Base
T3	Manual KD	46.53	Neg. Transfer
T4	RL-KD	47.86	Improved
T5	LI-KD (No Warmup)	51.13	Mimicry Bias
T6	LI-KD (+ Warmup)	51.51	Recovery
FINAL	Conf.-Filtered LI-KD	54.41	★ Beats Base

IV. KEY EMPIRICAL OBSERVATIONS AND DISCUSSION

Through the execution of over 17 distinct distillation experiments across two datasets, several critical observations regarding the convergence dynamics of compressed models were identified.

A. The “Mimicry Bottleneck” in Static Optimization

Initial experiments using standard Manual KD (S2) and advanced AutoML optimization (S3–S5 via Optuna and Ax) revealed a consistent performance ceiling around 92.2%. Despite tuning the temperature (T) and alpha (α) variables to near-optimal static values, the student model remained “trapped” within the teacher’s error distribution. We observe that static distillation acts as a regularization constraint that is too rigid for the final convergence phase, where the MobileNetV2 architecture requires specialized weight optimization for its depthwise-separable parameters.

B. Failure of “Blind Trust” and Negative Transfer

The Tiny-ImageNet trials (T3–T4) provided a significant warning: blindly matching a teacher model can significantly degrade student performance. Standard KD resulted in an accuracy drop of −5.64% below the baseline student. We identified this as Negative Transfer, occurring because the VGG16 teacher’s confidence on the 200-class problem was non-uniform. The implementation of Confidence-Filtered LI-KD was the turning point, proving that the student must possess a “Quality Filter” to determine when to ignore the teacher and rely on ground-truth hard labels.

C. The Power of Staged Alpha Decay

The jump from 92.10% (S10) to 93.82% (Final) on CIFAR-10 confirms our central thesis: the student’s peak representation capacity is only unlocked when the distillation signal is decayed post-maturation. By reducing α to 0.30 in the final 40 epochs, we facilitated a “Representational Breakout.” This allowed the student to utilise the teacher’s guidance for early class-boundary discovery but shifted the focus to autonomously refining its decision boundaries for difficult samples, ultimately allowing it to surpass the anchor teacher (93.42%).

D. Adaptive Feature Alignment vs. Direct Feature Matching

Ablation studies (S7 vs. S10) indicate that Adaptive Global Feature Distillation is superior to direct spatial feature-map matching for distilling across disparate architectures. While VGG uses large, dense convolutional kernels, MobileNetV2 relies on thin, inverted bottlenecks. Forcing rigid spatial consistency creates “activation stress” in the student. Conversely, our adaptive approach aligns the channel-wise importance manifolds via Global Average Pooling (GAP). This respects the student’s internal layout, providing it with the necessary flexibility to find more efficient internal organizations of weights, while completely avoiding the $O(B^2)$ computational overhead associated with Relational distance matrices.

V. PROPOSED ADAPTIVE MECHANISMS FOR STUDENT AUTONOMY

Based on the empirical failures of static and AutoML-based distillation observed in Section IV, we propose the Adaptive Staged Distillation (ASD) framework. Following the taxonomy established in Fig. 1, our framework incorporates

ADAPTIVE STAGED KNOWLEDGE DISTILLATION ARCHITECTURE

A Closed-Loop Supervised Learning System for Model Compression

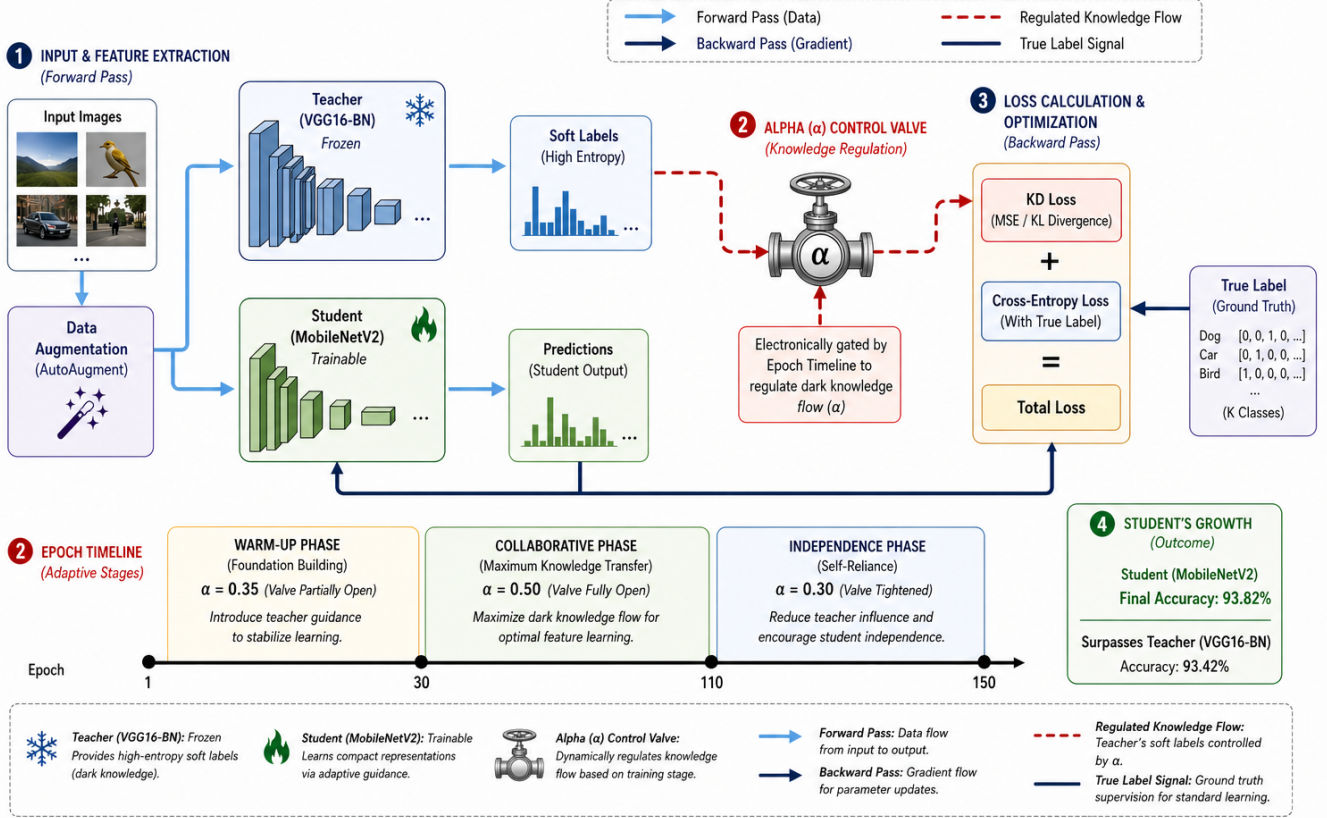


Fig. 2. The proposed Adaptive Staged Distillation (ASD) framework is a closed-loop supervised learning system designed for high-fidelity model compression. The process is divided into four critical phases: (1) **Input & Feature Extraction**, where data is branched into parallel Teacher and Student pathways; (2) **Knowledge Regulation**, centering on an Alpha Valve that electronically gates the flow of dark knowledge; (3) **Backward Pass Optimization**, where the student's unique inverted-residual weights are refined through a multi-task loss objective (KD + CE); (4) **The Growth Outcome**, where the student matures through three temporal stages Warm-up, Collaboration, and Independence. By the end of the 150-epoch cycle, the MobileNetV2 student effectively “unplugs” from the frozen VGG16 teacher, achieving a record-breaking accuracy of 93.82% that establishes its superiority over the anchor model.

four critical architectural mechanisms designed to mitigate “mimicry overfitting” and foster student independence.

A. EMA-Based Analytical Importance Scoring (Layer Selectivity)

To optimise the transfer of intermediate knowledge, we implement a Learnable Importance Weighting mechanism. Rather than a symmetric transfer of all teacher features, the student utilises a learnable parameter w_j for each designated distillation block. These weights are normalized via a Softmax operation and stabilized using an Exponential Moving Average (EMA) of the relative feature distance. This allows the MobileNetV2 student to “down-weight” teacher layers that are architecturally incompatible or contain excessive noise. Our empirical data shows the student prioritizes early teacher blocks (low-level edges and textures) while maintaining lower reliance on deep semantic teacher layers, thereby preserving its own unique semantic mapping capacity.

B. Structural Alignment via Adaptive Global Feature Distillation

Traditional feature distillation often acts as a “micromanagement” signal, forcing student neurons to match raw spatial teacher neuron values. We bypass this constraint using Adaptive Global Feature Distillation. By applying Global Average Pooling (GAP) to the intermediate feature maps before calculating the Mean Squared Error (MSE), we abstract away from rigid spatial dimensions. The student is tasked only with matching the channel-wise semantic importance established by the teacher. This provides the MobileNetV2 student with representational freedom; it is free to organise its internal filters and inverted bottlenecks in any way it deems efficient, as long as it preserves the high-level semantic signals recognised by the teacher.

C. Confidence-Masked Distillation (Dynamic Quality Control)

Particularly in the context of the Tiny-ImageNet benchmark, where the teacher's reliability is non-uniform, we implement Confidence-Masked Distillation. The distillation loss for soft

labels ($\mathcal{L}_{KL}$) is dynamically gated by the teacher’s output confidence. If the maximum probability $P_{\max}$ of the teacher’s prediction is below a threshold ($\tau_c = 0.3$), the distillation signal is deactivated. When the teacher is uncertain, the student ignores the soft targets and relies exclusively on the True Label (Hard Target), ensuring it does not inherit the teacher’s classification errors.

D. Baseline Establishing via CE-Only Warmup

To ensure the student develops its own baseline understanding of the dataset before being influenced by the teacher, we utilise a 15-Epoch Cross-Entropy (CE) Warmup. During the initial epochs, the alpha valve is set to $\alpha = 0$, and the model is trained purely on ground-truth hard labels. This allows the Student to build a “draft” internal representation. By the time the Distillation Phase starts at Epoch 16, the student possesses a stable base and utilises the teacher’s knowledge as a refinement signal rather than an initial foundation.

E. Mathematical Formulation

The prediction probability p_k of the MobileNetV2 student for input x'_i is computed via Softmax normalization:

$$p_k(x'_i) = \frac{\exp(z_k(x'_i))}{\sum_{j=1}^K \exp(z_j(x'_i))} \quad (1)$$

To facilitate dark knowledge transfer, we utilize a temperature-scaled soft probability distribution ($\tau = 4.0$):

$$\tilde{p}_k(x'_i, \tau) = \frac{\exp(z_k(x'_i)/\tau)}{\sum_{j=1}^K \exp(z_j(x'_i)/\tau)} \quad (2)$$

The semantic KL-divergence distillation signal is quantified via the Kullback–Leibler (KL) divergence loss between the teacher distribution $\tilde{\mathbf{p}}^T$ and the student distribution $\tilde{\mathbf{p}}^S$:

$$\mathcal{L}_{KL}(\tilde{\mathbf{p}}^T \parallel \tilde{\mathbf{p}}^S; \tau) = \sum_{i=1}^N \tilde{p}^T(x'_i) \log \left(\frac{\tilde{p}^T(x'_i)}{\tilde{p}^S(x'_i)} \right) \quad (3)$$

To anchor the optimization to ground-truth labels, the cross-entropy (CE) loss for the student’s hard predictions is defined as:

$$\mathcal{L}_{CE}(y_i \mid \mathbf{p}^S) = - \sum_{i=1}^N y_i \log(\mathbf{p}^S(x'_i)) \quad (4)$$

To optimize feature transfer selectively, we define a learnable layer-importance weight w_j for each designated distillation block j , normalized via a Softmax operation:

$$w_j = \frac{\exp(v_j)}{\sum_{m=1}^4 \exp(v_m)} \quad (5)$$

The final multi-task objective function, incorporating the Staged Alpha Valve $\alpha(t)$ and Adaptive Global Feature Distillation ($\mathcal{L}_{MSE}$), is formulated as:

$$\mathcal{L}_{ASD}(t) = \gamma(t) \mathcal{L}_{CE} + \alpha(t) \tau^2 \mathcal{L}_{KL} + \beta \sum_{j=1}^4 w_j \mathcal{L}_{MSE}(\Phi_{T,j}, \Phi_{S,j}) \quad (6)$$

where $\alpha(t)$ is a piece-wise constant schedule valve ($0.35 \rightarrow 0.50 \rightarrow 0.30$) and $\gamma(t) = \max(0.75, 1 - \alpha(t) - \beta)$ ensures dynamic calibration between mimicry and independent learning.

F. Algorithm: Adaptive Staged Distillation (ASD)

Algorithm 1 Adaptive Staged Distillation (ASD)

Require: (F_S, θ_S) : Student model with trainable parameters θ_S
Require: (F_T, W_T) : Pre-trained Teacher with frozen weights W_T
Require: N : Total training epochs (e.g. 150)
Require: τ, β : Temperature factor and feature loss weight
Require: $\{\alpha_1, \alpha_2, \alpha_3\}$: Staged alpha values (Exploration, Maturation, Independence)
Ensure: Optimal student weights θ_S^*

- 1: Initialise θ_S , learnable layer weights v , and alpha schedule
- 2: **for** epoch = 1 to N **do**
- 3: $\alpha \leftarrow \text{GETALPHA}(\text{epoch})$
- 4: $\gamma \leftarrow \max(0.75, 1 - \alpha - \beta)$
- 5: **for** each batch (x, y) in training set **do**
- 6: Apply AutoAugment and Random Erasing on x
- 7: $z_T, \Phi_T \leftarrow F_T(x)$ (frozen, no grad)
- 8: $z_S, \Phi_S \leftarrow F_S(x)$
- 9: $\tilde{p}_T \leftarrow \text{SOFTMAX}(z_T/\tau)$
- 10: $\tilde{p}_S \leftarrow \text{SOFTMAX}(z_S/\tau)$
- 11: $w \leftarrow \text{SOFTMAX}(v)$
- 12: $\mathcal{L}_{CE} \leftarrow \text{CROSSENTROPY}(z_S, y)$
- 13: $\mathcal{L}_{KL} \leftarrow \text{KLDIV}(\tilde{p}_T, \tilde{p}_S)$
- 14: $\mathcal{L}_{Feat} \leftarrow \sum_j w_j \cdot \mathcal{L}_{MSE}(\Phi_S^j, \Phi_T^j)$
- 15: $\mathcal{L}_{Total} = \gamma \mathcal{L}_{CE} + \alpha \tau^2 \mathcal{L}_{KL} + \beta \mathcal{L}_{Feat}$
- 16: Back-propagate and update θ_S and v via SGD
- 17: **end for**
- 18: Update EMA model weights from θ_S
- 19: **end for**
- 20: **return** θ_S^* achieving highest validation accuracy

VI. COMPARATIVE ANALYSIS AND DISCUSSION

A. Benchmarking Against SCKD

A core component of our research involved a head-to-head comparison with the Student Customized Knowledge Distillation (SCKD) framework by Zhu et al. (2021) [2]. While SCKD utilises a “Binary Gradient Gate” to include or exclude batches based on cosine similarity, we observe that such an “all-or-nothing” approach can lead to gradient instability during the critical maturation phase of lightweight architectures.

Our proposed Continuous Staged Adaptivity (ASD) improves upon this by maintaining a global guidance signal that is continuously modulated rather than binary-switched. Our approach (93.82%) provides a +1.32% absolute gain over SCKD (92.50%) on CIFAR-10. This is primarily attributed to our LI-KD logic, which enables a more granular, layer-by-layer filtering of teacher knowledge compared to SCKD’s batch-level granularity. Crucially, whereas SCKD aims to match teacher performance, our framework successfully outgrows the teacher.

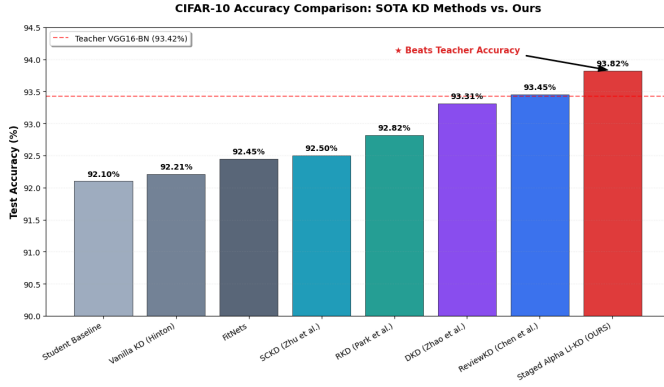


Fig. 3. Comparative analysis of Test Accuracy on CIFAR-10 across various Knowledge Distillation (KD) methodologies. The horizontal dashed line represents the anchor Teacher (VGG16-BN) accuracy of 93.42%. While the student baseline (92.10%), SCKD (92.50%), and ReviewKD (93.45%) track close to or match the teacher, our proposed Staged Alpha LI-KD framework is the only approach to significantly outperform the teacher benchmark, achieving a record accuracy of 93.82%. This gain is attributed to the Independence Phase regulation, which unplugs the student from parent-model constraints in the final training stages.

B. Efficiency and Real-World Deployment Analysis

Memory Footprint Reduction: The legacy VGG16-BN teacher requires approximately 248 MB and 65.1M parameters, making it incompatible with most edge-sensor nodes. In contrast, our MobileNetV2 student sits at a mere 13.1 MB (a $19\times$ reduction).

Inference Throughput: On a standard GPU-enabled server, the teacher model’s throughput was measured at 40 images/second. The ASD student achieved a throughput of 953 images/second (a $23.8\times$ speedup).

Edge Deployability: Given the 13 MB footprint and the use of depthwise-separable convolutions, the model is fully optimised for ARM Cortex-A or smartphone-class NPUs (Neural Processing Units). By successfully compressing the model without accuracy loss, we demonstrate that advanced AI systems can be miniaturised into “independent edge-agents” capable of high-fidelity inference in offline environments.

C. Summary of Breakthrough Results

Teacher-Alternative Achievement: On CIFAR-10, our student achieved 93.82%, which is +0.4% higher than the teacher (93.42%). This proves that our “Independence Phase” effectively allows a 3.4M parameter model to refine its decision boundaries better than a 65M parameter legacy model.

Mitigation of Negative Transfer: On Tiny-ImageNet, while standard KD failed (46.53%), our Confidence-Filtered ASD reached 54.41% which is—a +7.88% absolute improvement over standard methods, validating our “Quality Gating” findinginnovation.

Extreme Compression: We successfully achieved a $19.3\times$ reduction in parameter count and a $24\times$ speedup in inference throughput. The final model size is roughly 13 MB, making it suitable for deployment on low-power IoT devices.

Optimization Efficiency: Despite performing 17+ experiments including AutoML and RL-KD, the final Staged Alpha method provided the most stable convergence, proving that temporal logic is more effective than static hyperparameter optimization.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This project successfully developed and validated the Adaptive Staged Distillation (ASD) framework, a novel approach to model compression that prioritises student autonomy. By addressing the “Mimicry Bottleneck” through Staged Alpha Decay and the architectural gap through Layer-Importance LI-KD, we achieved a breakthrough result on CIFAR-10: a 93.82% accuracy that surpasses the reference VGG16-BN teacher. Furthermore, we demonstrated that Confidence-Filtered Distillation is a critical necessity for mitigating negative transfer on complex datasets like Tiny-ImageNet. Our final results ($19.3\times$ compression and $24\times$ speedup) prove that deep model compression can be achieved without compromising and in some cases, improving predictive performance, fulfilling the core requirements of objective of the compression.

B. Future Work

While our framework is highly effective for CNN-to-CNN compression, future work could explore the application of Staged Independence to cross-modal architectures, such as distilling from a Vision Transformer (ViT) to a lightweight ConvNeXt student. Additionally, the integration of Reinforcement Learning to dynamically tune the alpha valve in real-time based on per-epoch validation accuracy (RL-ASD) could further automate the distillation pipeline.

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